Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/29/2006
PI Name:	Kerkhof, Lee Ph.D.		
Project Title:	Characterization of Eubacteria in the Biological Water Processor (BWP)-Nitrogen Cycling		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08901	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
Start Date:	01/01/2001	**End Date:**	02/21/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pickering, Karen D. (Johnson Space Center)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Five bench scale biological water processors (BWP) at JSC have been investigated to monitor microbial population dynamics and gain insight into the microbial mechanisms controlling nitrogen transformations in an advanced life-support system (ALS). Primarily, the total, denitrifying, and nitrifying bacterial communities have been identified using molecular methods. Four experiments were conducted to determine: replicability in bioreactors, the role of inocula, the effects of system shutdowns on active bacteria, and the influence of loading on reactor performance. The overall objectives are to understand the microbial ecology of the bioreactors and provide small improvements in efficiency of resource recovery which may will translate to large reductions in equivalent system mass necessary to support a crew for long durations. The ability to monitor bacterial population changes may provide an early warning system to monitor degradation of the biological subsystems in the ALS far sooner than loss of function.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research proposed here will directly contribute to the design and optimization of the resource recovery sub-systems (RRS) or solid processing sub-systems (SPS). Since unrecoverable waste must be collected, stored, stabilized, for return to earth, small improvements in efficiency translate to large reductions in mass necessary for containment. Additionally, each pound of resource that can be successfully recovered further reduces the mass that must be lifted into orbit to sustain the crew. This is a significant reduction in ESM. Without information on the bacterial population inside the "black box" it become impossible to predict whether design improvements in single, lab-based studies can be successfully applied to biological hardware being designed for the International Space Station, a lunar colony, or a Mars mission. Finally, the insights gleaned from recycling graywater for a NASA spacecraft can be incorporated in wastewater systems and to recycle graywater for domestic use on Earth.
Task Progress:	Task Progress in Current Year: A. Replicability study 5 Denitrifying Bioreactors Researchers: L. McGuinness, L. Vega, K. Pickering, and L. Kerkhof Bioreactors hold great promise for treating graywater in an advanced life support system for space applications. However, questions remain regarding reproducibility and reliability of biological systems for long-term use. Although there have been numerous studies on ground-based biological systems, most studies focus on a single reactor or a simple (single carbon) waste stream. There have been very few studies on microbial communities in replicate reactors using a non-sterile, complex waste stream. In this report, we describe characterization of 5 replicate denitrifying reactors receiving a complex feed, including urine and limb washes from donors at Johnson Space Center over a 100-day period. Bacterial populations were tracked by characterizing 16S rRNA and nosZ genes by TRFLP to target the total and denitrifying microbial community. Results to date: The results demonstrated highly reproducible biofilms with nearly identical performance that slowly changed with time and exhibited highly predictable variability with respect to the bacterial community (TRFLP peak area) in all reactors. These results suggest that when designed for replication, bioreactors are not stochastic systems exhibiting chaotic behavior, but are biological systems that can be highly reproducible and reliable. A manuscript detailing the study is currently under review at Applied and Environmental Microbiology. B. Bromodeoxyuridine as a method for identifying active bacteria in bioreactors. Researchers: L. McGuinness, L. Vega, K. Pickering, and L. Kerkhof In order to address which bacteria are active within the bioreactor samples, we have recently used a DNA base analogue (bromodeoxyuridine; BrDU) that becomes incorporated into newly synthesized DNA during cell growth and division. This non-radioactive label has been shown to mimic the traditional thymidine uptake measurements in marine and estuarine systems with a nearly 1:1 correspondence over a broad range of activity levels (Steward et al, 1999). Additionally, the BrDU label can be detected and quantified immunologically using a protocol developed a decade ago (Kerkhof, 1992). In addition to the non-radioactive quantification ability, a mouse antibody for detecting BrDU can also be used in conjunction with magnetic beads to immunopurify the newly synthesized DNA for T-RFLP analysis (Borneman, 1999; Urbach et al., 1999). This allows for the rapid separation and identification of those bacteria that are replicating their genomes (growing) from the remaining portion of the microbial biomass that may be in stationary phase yet possess intact genomes. Results to date: The results from screening BWP samples at different levels within the reactors on day 120 reflect the history of various reactors during operation. The overall BrDU incorporation tracked with overall reactor performance. Specifically, reactor 4 produced the highest level of newly synthesized DNA and also has the highest level of TOC removal. Reactors 2 and 3 have nearly equal BrDU incorporation and similar TOC removal rates that are close to reactor 4. Reactor 5 showed less TOC removal and BrDU incorporation was not detected in the intermediate layers Reactor 1 experienced a low pH event shortly before sampling and BrDU incorporation was not detected at any level within the reactor at this time. A manuscript detailing the study is nearly ready for submission to Applied and Environmental Microbiology. C. A Perturbation Study of Replicate Bioreactors At the end of an experimental run using 5 replicate reactors, 2 reactors were taken offline for 24 hours and samples were collected for chemistry and microbiology. The reactors were re-started and the microbial response was assessed. While the reactors were quiescent, samples were taken for assessing which bacteria were induced and which were becoming inactive. Samples were collected and fingerprinted from total biomass and from the BrDU fraction. Results to date: Both quiescent reactors completely removed nitrate under the no-flow condition and much of the nitrite was removed as well. However, reactor 4 is producing a significant amount of nitrite after the pumps are turned back on while reactor 5 is better at converting the nitrate to an end product other than nitrite. This suggests that both denitrifying and nitrate respiring microorganisms are present in the reactors and that perturbations (such as the pH event) may be used to select for different microbial populations within the bioreactor. Additionally, we could demonstrate the level of BrDU incorporation rises dramatically when the reactor is not receiving gray water or is in recirculation mode. Furthermore, the numbers of different bacteria that respond to this anoxic condition also rises. Therefore, we believe that incorporating a brief period of no-flow on the reactors will induce many members of the microbial community that are growing slowly in the reactor. A manuscript detailing the study is also being prepared for submission to Applied and Environmental Microbiology.
Bibliography Type:	Description: (Last Updated: 09/21/2006)
Articles in Peer-reviewed Journals	McGuinness LM, Salganik M, Vega L, Pickering KD, Kerkhof LJ. "Replicability of bacterial communities in denitrifying bioreactors as measured by PCR/T-RFLP analysis." Environ Sci Technol. 2006 Jan 15;40(2):509-15. PMID: 16468396 , Jan-2006
Papers from Meeting Proceedings	Vega L, Kerkhof L,McGuinness L, Pickering KD. "Evaluation of Performance of Five Parallel Biological Water Processors." 34th International Conference of Environmental Systems. Colorado Springs, Colorado, July 2004. SAE paper 2004-01-2515. 34th International Conference of Environmental Systems. July 2004 , Jul-2004

Presentation	McGuinness, L., K.D. Pickering, L. Vega, and L. Kerkhof. "Molecular characterization of N-cycling bacteria in the BWP. " Conference on Space Habitation Research and Technology Development. Orlando, Florida. Jan-2004
Presentation	L. Vega. , L. Kerkhof, L McGuinness, and K.D. Pickering. "Evaluation of Performance of Five Parallel Biological Water Processors. SAE 200-01-2515. " 34th International Conference of Environmental Systems. Colorado Springs, Colorado. Jul-2004